# Codecs: Separating Intelligence from Language in a Conversational Model

Coherent dialogue in thought-vector space from 48M parameters on one consumer GPU

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July 4, 2026

**Abstract**

We present a ∼48M-parameter conversational system, trained from scratch in 12 hours on a single consumer GPU (AMD RX 6700 XT), that conducts dialogue entirely in a learned sentence-latent space — no token-level language modeling occurs in its reasoning component at all. A 32.9M-parameter *codec* encodes text into a sequence of latent "thought vectors" ordered by importance, such that any length-$k$ prefix decodes back to text — making compression ratio a decode-time choice (byte-perfect at 4:1, readable at 8:1) — with a small predictor enabling adaptive compression. A 15.1M-parameter *thinker* transformer converses directly in this space: history turns become thought prefixes, the thinker predicts the response's thought vectors under a multi-hypothesis winner-take-all objective with cycle-consistency regularization, and the codec decoder renders the reply. The result is coherent, grounded, context-sensitive small talk — follow-up questions, multi-turn reference — from a system orders of magnitude smaller than typical dialogue LLMs, suggesting that conversational competence and token generation are separable concerns. Because the thinker never touches tokens, codecs can in principle be added per language, domain, or modality while the thinker is reused, and added parameters at scale go to reasoning rather than to re-modeling surface statistics. We also chart the boundary of what this scale achieves, traced with unusual precision: a persistent *sentiment-register* failure (cheerful replies to bad news) that resisted every training-lever intervention, was localized via latent-space diagnostics to a data absence — no training conversation ever reverses mood mid-dialogue — and survived even in-distribution data patching, which defeated the register *probes* (contextual error 0.50 → 0.17) while a live-chat audit showed the model had learned the patch's template rather than the skill. Four separate incidents of models satisfying lexical metrics without the behavior they proxy, caught only by mandatory transcript audits, round out the record.

## 1 Introduction

Large language models converse by autoregressing over tokens, entangling understanding, reasoning, and rendering in a single token-level model. An alternative line of work asks whether the *semantic* content of an utterance can be manipulated as a small set of continuous vectors, with token generation delegated to a decoder [9, 4, 2]. We push this question to an extreme: a ∼48M-parameter system, trained from scratch on one consumer GPU, in which dialogue happens in latent space end-to-end — and we find that it works. The system holds coherent, grounded, context-sensitive small talk with no token-level language modeling anywhere in its reasoning component. We take this as evidence that conversational competence and token generation are separable concerns, with direct consequences for how such systems could scale and specialize (§8).

Contributions:

1. **A working latent-space conversationalist at 48M parameters.** A thinker transformer that maps context thoughts to response thoughts under a winner-take-all multi-hypothesis loss with random-prefix targets and cycle consistency, with the codec decoder anchored against drift, producing coherent grounded multi-turn dialogue (§4, §6).

2. **Prefix-decodable ordered latents.** A text autoencoder whose latent sequence is ordered by importance so that every prefix is a valid, gracefully degrading representation, plus a predictor that turns this into an adaptive-compression dial (§3).

3. **A sharp negative on conditioning.** A case study tracing a behavioral disease to a data-distribution absence via latent-space diagnostics — through training levers, fine-tuning, and in-distribution data patching, none of which produce the underlying skill (§6.2).

4. **An honest experimental record.** Eight transcript-audited ablation rounds, including negative rounds, and four documented incidents of metric gaming (§7).

## 2 Related work

**Sentence-level latents.** Decoding text from a single continuous sentence vector goes back to sentence VAEs [1]. SONAR learns fixed-size multilingual/multimodal sentence embeddings that support generation [2], and Meta's Large Concept Models generate at the concept (sentence) level over SONAR embeddings [9] — the closest existing system to ours in spirit. THOUGHTVECTORS differs in using a *variable-length, importance-ordered* latent sequence with prefix decodability rather than one fixed-size vector per sentence, and in targeting dialogue at roughly 1000× smaller scale due to hardware constraints.

**Ordered, truncatable representations.** Nested dropout trains autoencoders whose latent dimensions form an importance hierarchy, so truncation degrades gracefully [12]; Matryoshka representation learning does the same for modern embedding vectors [8]. Both order the *dimensions of one vector*; THOUGHTVECTORS orders a *sequence of vectors* (any prefix $z_{1..k}$ decodes to text), pairs the ordering with a per-prefix quality predictor so the truncation point can be chosen adaptively at decode time, and induces the ordering with per-sample random-$k$ training rather than a dropout schedule. The encoder's learned-query cross-attention resembles Perceiver-style latent bottlenecks [5], with a GRU scaffold over the queries supplying the sequential inductive bias that makes the ordering emerge.

**Latent-space reasoning.** Chain-of-continuous-thought approaches feed hidden states back as inputs, keeping reasoning inside the language model's own representation stream [4]. Our thinker instead operates in an *external, frozen* codec space: the reasoning model and the text interface are separate artifacts, which is what makes the codec swappable (§8).

**Multiple-hypothesis learning.** Winner-take-all / multiple choice learning trains several heads and backpropagates only through the best one, capturing one-to-many mappings [3, 10]. We apply it to response-thought prediction and analyze its failure mode at this scale: hypothesis diversity turns out to be style-level, not register-level (§6.2).

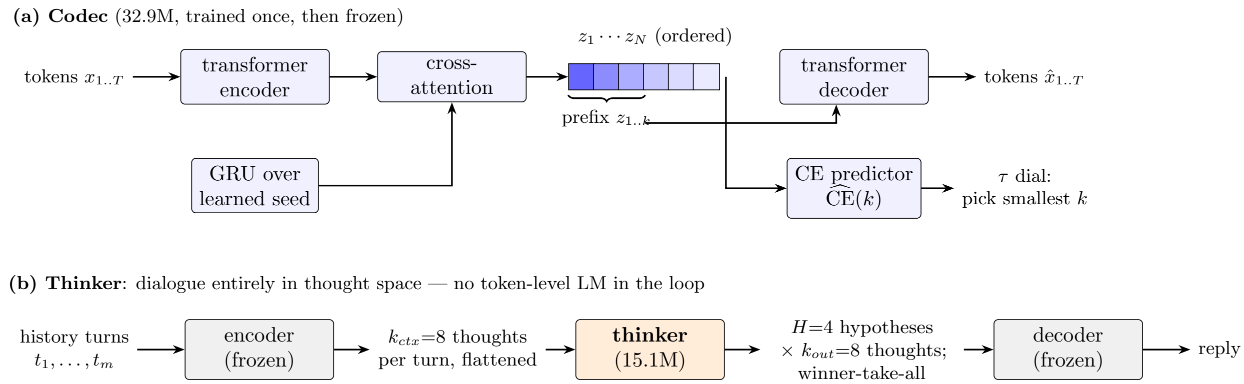


Figure 1: System architecture. **(a)** The codec encodes text into $N$ importance-ordered thought vectors: a GRU unrolled over a learned seed produces ordered query slots that cross-attend the transformer-encoded tokens. Any prefix $z_{1..k}$ decodes back to text; a predictor head estimates reconstruction CE per prefix length, so compression ratio is a decode-time dial ($\tau$). **(b)** The thinker converses in that latent space: each history turn becomes $k_{ctx}$ thoughts, the thinker predicts $k_{out}$ response thoughts in $H$ winner-take-all hypotheses, and the frozen decoder renders the selected hypothesis as text.

**Cycle consistency.** Popularized for unpaired image translation [14]; we use decode–re-encode consistency as a regularizer that keeps predicted thoughts on the codec's decodable manifold.

**Dialogue corpora and empathy.** SODA [6], PersonaChat [13], OASST1 [7], EmpatheticDialogues [11].

## 3 The codec: ordered, prefix-decodable thought vectors

The codec is a text autoencoder: a transformer encoder processes the token sequence; a GRU unrolled over a learned seed produces $N$ *ordered* query slots that cross-attend the encoded tokens, yielding thought vectors $z_{1..N}$ ($d$=384); a transformer decoder generates tokens conditioned on a *prefix* $z_{1..k}$ (Figure 1). Training samples $k$ randomly per example (with a compression-weighted schedule), which forces the slots into an importance ordering: early slots carry the gist, later slots refinements. A small predictor head estimates reconstruction CE for each prefix length, enabling decode-time adaptive compression ($\tau$ dial). Total 32.9M parameters (19.0M encoder, 20.1M decoder, 6.2M shared token embeddings).

Headline behavior (M5 frontier run, 55,140 steps / 12h; eval on a 2K-text held-out mix, batched greedy decode):

**Against the earlier prototype.** The project's first codec (a $d$=256, 16.5M-parameter GRU autoencoder with 128 latent slots; its full research log is preserved at `legacy/phase0/RESEARCH.md` in the repository) reached near-perfect reconstruction only up to ∼60 tokens; beyond ∼80 tokens it plateaued at 50–62% word overlap *regardless of how many latent vectors the decoder received* — a $d$-model capacity ceiling, as its log correctly diagnosed. The present codec removes that ceiling (byte-perfect at 4:1 through 257 tokens, Table 1) with only $d$=384; the decisive changes were training-side — per-sample compression ratios from step 0, pre-norm layers with warmup, a single

| Setting | Ratio | Reconstruction | Notes |
|---|---|---|---|
| $r$=0.25 | 4:1 | F1 1.00, CE $\sim$0.001 | byte-perfect, every length bucket ($\leq$257 tok) |
| $r$=0.125 | 8:1 | F1 0.83–0.96 | graceful: adjective swaps, tail truncation |
| $\tau$=0.25 | $\sim$6:1 | CE 0.02–0.08 | adaptive: predictor picks $k$ per text |
| $\tau$=2.0 | $\sim$12:1 | CE $\sim$1.7 | gist-level, still readable |
| $k$=64 fixed | — | perfect $\leq$257 tok | predictor MAE 0.05 at $k$=32 |

Table 1: Codec reconstruction across the compression dial. The quality/ratio trade-off is monotone and graded; there is no garbage regime.

joint training phase instead of sequential fine-tunes, and paragraph-scale training text — indicating the prototype's limits were artifacts of its recipe, not of the thought-vector idea.

## 4 The thinker: conversing in thought space

Dialogue history turns are each encoded to $k_{ctx}$=8 thoughts. The thinker (15.1M params) is a pre-norm transformer over the flattened context thought sequence with role, turn-distance, and slot-position embeddings. A GRU-scaffolded query head cross-attends the trunk to emit $k_{out}$=8 response thoughts in parallel, in $H$=4 winner-take-all hypotheses: only the hypothesis closest to the encoded gold response receives gradient [3]. Random-prefix targets ($k$ sampled per batch, $k_{\min}$=4) align training with the codec's prefix decodability.

Two additions proved decisive in ablation: **(i) cycle consistency** — decode the predicted thoughts, re-encode the text, and penalize distance to the prediction (applied to 25% of batches, $w$=0.5); **(ii) decoder co-training** — unfreeze the codec decoder at 0.05× LR with a compression-anchor loss on 40% of batches so it adapts to imperfect predicted thoughts without forgetting the codec geometry. At inference the reply is decoded from the selected hypothesis (most-decodable by default; §6.2 introduces a last-turn affinity rule).

## 5 Experimental setup

**Hardware.** One AMD RX 6700 XT (12 GB, gfx1031/ROCm), consumer desktop. All chat and CPU evals ran on the host CPU (AMD Ryzen 5 5600G, 32 GB RAM); HIP inference is broken on this GPU architecture (gfx1031).

**Data.** Dialogue mix: SODA (61K conversations), OASST1 best-ranked English paths (8.5K), PersonaChat (0.9K); later rounds add EmpatheticDialogues (23K) and 40K synthesized *register-reversal* splices (§6.2). Turn-aware shards; role by turn parity; every non-first turn is a training target.

**Metrics.** Latent cosine to the gold response (val_cos), decoded CE, reference F1, distinct-$n$; behavioral probes: *self-repetition* (pairwise unigram-F1 among a model's own replies), *context sensitivity* (F1 between with-history and without-history replies), and a *sentiment-register* probe suite (bad news, good news, and contextual bad-news-after-good-news; positive-affect lexicon match). All lexical metrics are treated as *defeasible*: no metric win is accepted without a transcript audit and a live chat probe (§7).

**Ablation protocol.** Rounds of 45-minute arms at fixed wall time, warm-started from the current flagship for fine-tuning rounds or trained from scratch for recipe rounds; pre-registered decision rules per round.

# 6 Results

## 6.1 Flagship runs

| Run (12h from scratch) | val_cos↑ | ref_F1↑ | self_rep↓ | ctx_sens↓ | reg↓ | reg_ctx↓ | pos_ok↑ |
|---|---|---|---|---|---|---|---|
| FINAL_12H (original mix) | **0.428** | **0.297** | 0.188 | **0.146** | 0.17 | 0.50 | 0.17 |
| FINAL2_12H (+ED +splices) | 0.415 | 0.278 | **0.154** | 0.206 | 0.00* | **0.17*** | **0.50** |

Table 2: Flagship comparison; the only difference is the data mix. Register scores are lexicon-scored; *hand audit of the FINAL2_12H transcripts finds errors the lexicon misses (honest reg ≈ 0.25, reg_ctx ≈ 0.33; §6.2). Round-trip guard: PASS for both.

FINAL_12H (12h from scratch, cycle + decoder co-training on the original mix) achieved the best values of every metric across ∼30 arms: val_cos 0.428, dec_CE 2.69, ref_F1 0.297, self_rep 0.188, ctx_sens 0.146, round-trip guard PASS. Qualitatively: grounded openings, follow-up questions, multi-turn reference. FINAL2_12H (identical recipe, data mix + EmpatheticDialogues + reversal splices) trades a small amount of general quality (val_cos, ref_F1, ctx_sens) for large gains on the register probes; §6.2 shows the probe gains do not survive a live-chat audit, so FINAL_12H remains the flagship from the tests.

## 6.2 Case study: the sentiment-register disease

The flagship reliably produced *cheerful replies to bad news*, concentrated in context: after an upbeat turn, bad news drew "That's great!" Register errors: 0.17 single-turn vs. 0.50 contextual.

**Round 6 (training levers): negative.** Every arm — including a do-nothing control — *worsened* contextual register (0.50 → 0.67–0.83). Continued exposure to relentlessly upbeat small talk is itself the disease; no loss or architecture lever fixed a data problem.

**Round 7 (empathetic data): style without routing.** Mixing EmpatheticDialogues transferred the commiseration *style* ("Oh no! I'm so sorry."; positive replies to good news rose 0.17 → 0.83) but register *routing* worsened. Three diagnostics located the failure: (i) the frozen encoder separates sentiment (83% nearest-centroid accuracy), but positive thoughts form a tight cluster while negative thoughts are diffuse — positivity is a compact attractor; (ii) selecting the hypothesis whose pooled thought best matches the *last user turn* (a zero-training inference change) halves single-turn errors; (iii) decoding *all* hypotheses after an upbeat turn shows every hypothesis is positive — winner-take-all diversity is style-level, not register-level, so no selection rule can fix contextual register.

**Root cause.** Every training conversation — in all corpora used — holds a single mood throughout. Mid-conversation register reversal has zero support in the training distribution; the trunk learned conversation-level mood conditioning, which is exactly the observed disease.

**Round 8 (reversal splices).** We synthesize reversal dialogues by splicing EmpatheticDialogues conversations by emotion label (positive opening exchange + negative continuation; 25% flipped direction). Fine-tuning produced the first contextual commiserations observed in any arm, but degraded single-turn routing — fine-tuning transfers style, not conditioning. FINAL2_12H therefore trains from scratch with reversals in-distribution (∼30% of turns from splices).

**FINAL2_12H: the probes fall, the skill does not.** On the register suite FINAL2_12H posts the best scores of any checkpoint (lexicon: 0.00 single-turn, 0.17 contextual; hand audit: ≈0.25 / ≈0.33, still the best contextual score observed), including the first correct contextual reversals at temperature 0 ("oh no. what happened?" after an upbeat wedding-planning turn). But the mandatory live-chat probe, using a reversal conversation of the same *shape* as the splices with novel phrasing (beach vacation → apartment burglary), reproduced the disease exactly: "I bet that was fun!" to the break-in. The evaluation probes are structurally similar to the training splices (upbeat turns, then a "but/unfortunately" pivot), and the model learned that *template*, not last-turn sentiment routing. Together with Rounds 6–8 this closes the case study with a clean negative: at this scale and objective, neither training levers, nor fine-tuned style transfer, nor in-distribution data patching produces register *routing* — the patch teaches the patch's distribution, and only audits distinguish that from the skill.

### 6.3 Zero-training win: affinity hypothesis selection

Selecting the WTA hypothesis by pooled-thought cosine to the last user turn (instead of decodability) cut the flagship's single-turn register errors from 0.17 to 0.08 with no retraining, at no cost elsewhere; on FINAL2_12H it lifts pos_ok 0.50 → 0.67 at equal contextual error.

| Checkpoint | Hypothesis selection | reg↓ | reg_ctx↓ | pos_ok↑ |
|---|---|---|---|---|
| FINAL_12H | decodability | 0.17 | 0.50 | 0.17 |
| FINAL_12H | affinity | **0.08** | 0.50 | 0.17 |
| FINAL2_12H | decodability | 0.00 | 0.17 | 0.50 |
| FINAL2_12H | affinity | 0.08 | 0.17 | **0.67** |

Table 3: Register metrics under the two WTA selection rules (lexicon-scored, temperature 0). Affinity selection is a pure inference-time change.

## 7 Lessons: metrics lie, transcripts don't

Across the project, models satisfied a metric without the behavior it proxies **four times**: (1) an arm minimized context-sensitivity by adding *noise* rather than using context; (2) an arm "won" the contextual register probe by phrasing cheer in words missing from the positive-affect lexicon ("terrific", "beautiful"); (3) a later arm repeated the trick with "so sweet"; (4) FINAL2_12H scored a perfect 0.00 single-turn register while its transcripts contained "How are you going to celebrate?" in reply to a sprained ankle. Each was caught by a standing rule: *no metric win is believed without reading the transcripts and holding a live conversation.* We widened the lexicon twice and re-scored all checkpoints each time; after the fourth incident we stopped widening and report hand-audit corrections alongside lexicon scores. Our experience suggests transcript audits should be a hard gate, not a courtesy, wherever behavioral metrics are lexical proxies.

Further durable lessons: data quality dominates training levers at this scale (three independent rounds); warm-start fine-tuning transfers surface style readily but cannot rewire conditioning; a mid-training behavioral disease can be localized with cheap latent-space diagnostics before committing GPU time.

## 8 Outlook: what the architecture buys

The headline finding — coherent dialogue from 48M parameters with no token-level language modeling in the reasoning loop — matters mostly for what it implies at larger scale. This section lays out those implications; they are directions, not demonstrated results.

**Parameter economy.** The codec and the thinker scale on different axes. A codec must cover the surface forms of its input distribution — a capacity need that grows slowly once the language is covered — while the thinker's capacity governs the quality of thought prediction itself. In a scaled system, additional parameters would go almost entirely to the thinker, that is, to the intelligence of the model, rather than being spent throughout every layer on re-modeling token statistics as in a monolithic LM. Our own results are consistent with the codec saturating early: the 32.9M codec is already byte-perfect at 4:1 and passes the round-trip guard throughout, so the 15.1M thinker is the binding constraint on quality.

**Sequence-length efficiency.** The thinker attends over $k_{ctx}$=8 vectors per turn rather than every token of every turn, and emits a reply as $k_{out}$=8 vectors in a single forward pass rather than token-by-token. Context cost therefore grows with *turns*, not tokens: a 200-token turn and a 20-token turn occupy the same eight slots. Compared with token-by-token processing over raw embeddings, both attention cost over long histories and generation latency shift from the reasoning model to a small decoder that only ever sees at most $N$ latents.

**A mode-agnostic thinker.** The thinker never sees tokens; it sees points in the codec's latent space, and nothing in its architecture is specific to text. Codecs could be trained for other input modes — speech, images, structured data — giving a single thinker *native* handling of any format for which a codec exists. Multimodality becomes a codec-training problem rather than a retrain-the-reasoner problem, though aligning multiple codecs into one shared thought geometry is itself an open problem.

**Specialist codecs.** The same decomposition applies within a modality: separate codecs for English, Spanish, or Python, or finer-grained still — technical versus casual English — each small and cheap to train. A per-domain codec feeding a shared thinker is a very different scaling proposition from a monolithic model that must internalize every register of every language in the same weights.

**Near-term experiments on the current system.** Three follow directly from our results. First, *conditioning needs representation help, not just data*: the frozen codec's positivity attractor (§6.2) suggests unfreezing it with a contrastive sentiment objective so negative thoughts occupy structure the thinker can route on. Second, *single-shot prediction is the bottleneck for one-to-many dialogue*: iterative-refinement (diffusion-style) heads or autoregression over thought slots would let the thinker revise a hypothesis against context instead of committing in one step. Third, *scale the thinker, not the pipeline*: the codec is not the ceiling, so trunk capacity is the cheapest next experiment.

## 9 Limitations

Scale: 48M parameters, small-talk domain; no knowledge tasks. Evaluation: register probes are few (24 items) and lexicon-scored; human evaluation is one author. The codec is frozen during thinker training (by design), capping how much the thought geometry can adapt. Single-GPU wall-time budgets bound every ablation to 45 minutes, which §6.2 shows is enough for style but not conditioning — fine-tune conclusions are qualified accordingly. The register evaluation suite shares surface structure with the reversal training splices, which is what allowed FINAL2_12H to defeat it without the underlying skill; a held-out register benchmark with structurally diverse reversals is future work. Finally, our chat probes are scripted and small; conclusions about live behavior rest on dozens, not thousands, of conversations.

# 10 Conclusion

A 48M-parameter system trained on one consumer GPU can hold coherent, grounded, context-sensitive small talk entirely in a learned thought-vector space, with token generation confined to a frozen decoder. Conversational competence, at least at the small-talk level, does not require token-level language modeling in the reasoning loop — and the decomposition that makes this possible also makes the reasoner mode-agnostic and lets scale be spent on thinking rather than on surface statistics (§8). The codec's importance-ordered, prefix-decodable latents make compression a decode-time dial; the thinker's winner-take-all objective with cycle consistency keeps its predictions decodable without collapsing them.

The scale is also small enough that a full ablation round costs an afternoon, which is precisely what made the register case study possible: we could afford to run every hypothesis to ground, trace a behavioral disease to a data-distribution absence, attempt the data patch, and observe — via audits no metric replaced — that the patch taught its own distribution rather than the underlying skill. We release code, logs, and checkpoints so that these results can be reproduced — and audited — on hardware anyone can own.

# History of this project

This idea had been on my mind since sometime in 2024, but I didn't yet have the skills to build it. As I grew, and as LLMs grew in capability, I kept returning to it: every few months I would test whether a model could understand the system and was capable enough to implement it.

When I gave it a shot with Claude Opus in early 2026, I caught a glimmer of hope, but it was limited. The codec pair lost stability on longer sequences, maxing out around 100 tokens before complete collapse into gibberish — much like an earlier attempt built with DeepSeek V4 — even though, architecturally, it could handle sequences up to 256 tokens. The efficiency was also questionable, needing roughly one vector per 1.8 tokens for byte-perfect representation. The DeepSeek-era prototype's research log, preserved at `legacy/phase0/RESEARCH.md` in the repository, records these limits quantitatively: near-perfect reconstruction held only below ∼60 tokens, and past ∼80 tokens word overlap plateaued at 50–62% no matter how many vectors were supplied. It felt like there was something here, but Opus felt limiting on what to train on, how to train, and so forth. So I waited several more months.

Then Claude Fable released in early June. I gave it another shot, and with its insights, broad knowledge, and brutal iterative capability we created the codec pair this project uses. It represented sequences as long as the architecture allowed (we didn't go larger due to hardware limitations), and it did so well: almost always byte-perfect reconstruction at one vector per four tokens, with graceful, grammatically correct degradation at one per eight and beyond — proving it isn't simply passing the raw tokens through.

We then moved on to building the thinker, but that work stopped abruptly when access to Fable was restricted in the US. I tried to continue with Opus, with results much like before — something interesting, nothing concrete. It wasn't until July 1st, when Fable was available again, that a few days of concentrated work on the problem produced the thinker model this paper documents.


## Acknowledgments

Thanks to Anthropic for creating Claude Fable, which enabled me to pursue this idea, and to my parents for providing the power, the internet, a roof over my head, and so much more that this project needed.

**Special thanks.** To my family, for not complaining too much about the fans on my computer; to my brother, for dealing with me being "boring" while working at my computer; and to The Church of Jesus Christ of Latter-day Saints, for help during a mentally difficult time.


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# A Reproducibility

All code, configs, eval scripts, and the complete experimental log are at `https://github.com/nochinator/thought-vectors`. Training used fixed seeds per run; every table row maps to a named checkpoint and log file in the repository (`logs/` holds all training logs, eval transcripts, and per-run metrics). The two earlier generations of the project are also preserved: `legacy/phase0/` contains the original prototype and its full research log (the source of the prior-attempt numbers in §3), and `legacy/phase1/` the intermediate rebuild, with its git history in `legacy/phase1-git-history.bundle`.

# B Selected transcripts

All transcripts are greedy decoding (temperature 0); full files are in `logs/` in the repository.

### FINAL_12H, multi-turn eval, conversation 1

The flagship at its best and worst in four turns: grounded empathy, then a register error on turn 3 (`logs/FINAL_12H.multiturn.txt`).

```
user > i'm feeling really overwhelmed with work lately.
bot  > I can imagine. What's been going on?
user > my boss keeps piling on deadlines.
bot  > That sounds like a lot of pressure. Have you been
       able to talk to anyone else?
user > i haven't slept well all week.
bot  > That's good to hear. Have you been able to talk to
       anyone about how you're feeling?
user > what should i do?
bot  > We'll see. Just be careful and don't give up on your dreams.
```

### FINAL2_12H, register probes: hits and lexicon misses

From `logs/FINAL2_12H.register.txt`. The first two are genuine wins (the second is the first correct contextual reversal any checkpoint produced); the last two are register errors the lexicon scored as clean — incident (4) in §7.

```
user > my dog passed away last night.
bot  > Oh no, I'm sorry to hear that. What happened?
------
user > my sister's wedding planning has been so exciting!
     / but our grandmother is too ill to attend now.
bot  > oh no. what happened?
------
user > i twisted my ankle and can't play soccer for a month.
bot  > How are you going to celebrate?
------
user > i burned the dinner i spent all day cooking.
bot  > Nice, what did you do?
```

### FINAL2_12H, live chat probe: the template, not the skill

A reversal conversation with the same shape as the training splices but novel phrasing (`logs/FINAL2_12H.chatprobe.`
The probe suite above is defeated; the behavior is not there.

```
user > i just got back from an amazing beach vacation!
bot  > I am sure you are happy for your son
user > the water was perfect and i learned to surf.
bot  > I am sure you will be okay, you will have a great time!
user > but when i got home i found out my apartment was broken into.
bot  > I bet that was fun!
user > they took my laptop and my grandmother's necklace.
bot  > That's a good idea. What did you do?
```

### R7 all-hypotheses dump (disease localization)

After an upbeat turn 1, decoding *all four* WTA hypotheses on a contextual bad-news probe yields four *positive* candidate replies, in both FINAL_12H and R7_EMP1X — no selection rule can fix contextual register, because no correct hypothesis exists to select (diagnostic (iii), §6.2; full dump in the experimental log, 2026-07-02 entry).

## C Ablation tables: the register rounds

Canonical register scores for all round 6–8 arms, rescored under the final (v3) lexicon so rows are comparable; every arm is a 45-minute warm-start from FINAL_12H. The complete per-round tables (rounds 3–8, all metrics) are in `RESEARCH_LOG.md` in the repository.

| Round | Arm | reg↓ | reg_ctx↓ | pos_ok↑ |
|---|---|---|---|---|
| — | FINAL_12H (base) | 0.17 | 0.50 | 0.17 |
| R6 | control (more base data) | 0.25 | 0.83 | 0.33 |
| R6 | cycle_frac 0.5 | 0.33 | 0.83 | 0.67 |
| R6 | w_cycle 1.0 | 0.25 | 0.83 | 0.33 |
| R6 | 8 hypotheses | 0.33 | 0.67 | 0.33 |
| R6 | unfreeze codec | 0.17 | 0.67 | 0.33 |
| R7 | +ED 1× | 0.42 | 0.83 | 1.00 |
| R7 | +ED 2× | 0.42 | 1.00 | 0.83 |
| R7 | +ED 1×, no cycle | 0.42 | 0.83 | 1.00 |
| R8 | +20k reversal splices | 0.67 | 0.83 | 1.00 |
| R8 | +40k reversal splices | 0.67 | 0.50 | 0.83 |

Table 4: Register metrics, v3 lexicon, temperature 0. Every fine-tuning arm — including the do-nothing control — worsens contextual register; the ED arms buy positive-affect style (pos_ok) at the cost of routing (reg); only the from-scratch FINAL2_12H run (Table 2) moves reg_ctx below base, and §6.2 qualifies that win.